HAWK Fakultät Ingenieurwissenschaften und Gesundheit Göttingen

POLITECNICO MILANO 1863

Repetition to

ADVANCED TOPICS IN ENGINEERING PHYSICS

'Plasma and its applications'

including answers

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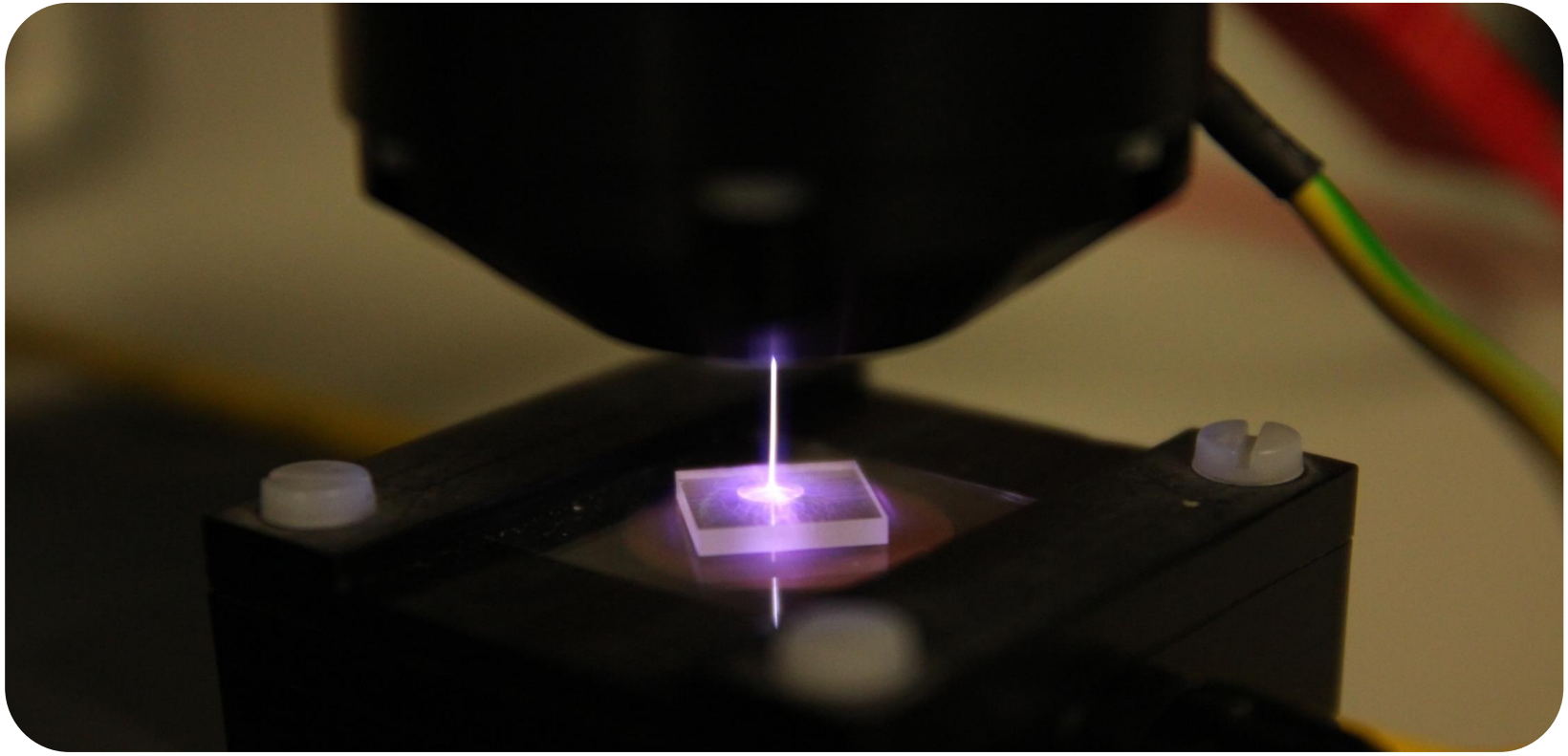
2 Plasma discharges and sources



- 46. Name at least three effects that may cause deviations from thermodynamic equilibrium within plasmas.
  1. The statistic sum of all collisions and energy absorption in the electric field results in electron energy distribution function EEDF (There is not the one and only electron energy!)
  2. The mean electron energy is higher than the mean excitation energy of heavier particles.
  3. The mean electron temperature is higher than the translation temperature (= ‘true gas temperature’) of heavy particles.
  4. The vibrational temperature is higher than the rotational temperature.

- 47. In which case does the electron energy distribution function not correspond to the Maxwell distribution?
  - In case of low electron density in comparison to gas density (for low degrees of ionisation, thus).

- 48. What is the basic assumption of the so-called Druyvesteyn distributio?
  - The loss or gain of energy via collisions with the background gas is described by a constant, electron-energy-independent cross section. It is valid for non-equilibrium plasmas.

- 49. What is the reason for the extremely non-uniform population of energy levels in non-equilibrium plasmas?
  - The excitation cross sections of the energy levels differ (quite significantly).

- 50. What is an optically ‘thick plasma’?
  - Within an optically thick plasma, the absorption distance is much smaller than the distance to the edge of plasma ($s_{abs} << L$)

- 51. Name an example of a plasma, where the mean electron temperature is higher than the translation temperature of heavy particles.
  - Dielectric barrier discharge (DBD) plasma

- 52. What is the ‘translation temperature’ of a plasma?
  - The translation temperature is the ‘true gas temperature’ of a plasma. Thus, it corresponds to the temperature of the neutrals an –as an approximation – of ions.

- 53. Why do plasmas feature higher vibrational temperature than rotational temperature?
  - The vibrational temperature is higher than the rotational temperature the since excitation cross sections of vibrational states are usually higher than for rotational states. Thus, more vibrational states than rotational ones are initiated.

- 54. Name some loss processes that may occur within a plasma.
  - Heat transfer and accompanying cooling
  - Photo emission
  - Loss of charge carriers
  - Changes in chemical
  - Plasma composition

- 55. Name the three most important types of DC plasma discharges
  - Townsend discharge, glow discharge, arc discharge

- 56. Give a selection of – about three – parameters that can be used for the classification of plasmas.
  - Pressure range, type of excitation (DC, AC, RF…), temperatures

- 57. What is the main characteristics of isothermal plasmas?
  - The degrees of freedom of all types of particles (neutrals and charged ones) feature a common temperature…as an approximation.

- 58. What is a Townsend discharge?
  - A non-self-maintaining discharge where free electrons are generated via avalanche ionisation

- 59. Name two mechanisms, which lead to the generation of the initial electron of a Townsend discharge.
  - Ionisation of gas atoms or molecules by incident cosmic irradiation.
  - Release of electrons from solid surface via photoelectric effect.

- 60. Which characteristic parameter describes the number of ionisation events per electron and distance?
  - The first Townsend coefficient α

- 61. How to determine the energy of an electron between two collisions?
  - It is given by the product of the free mean length of path and the electric field strength.

- 62. What are the main influencing parameters for the determination of the number of electrons generated via avalanche ionisation?
  - Electric field strength (= ‘acceleration’) and
  - Pressure (= particle density and thus free mean length of path)

- 63. What is the so-called ‘reduced field strength’?
  - The reduced field strength is the ratio of the electrical field strength and particle density.

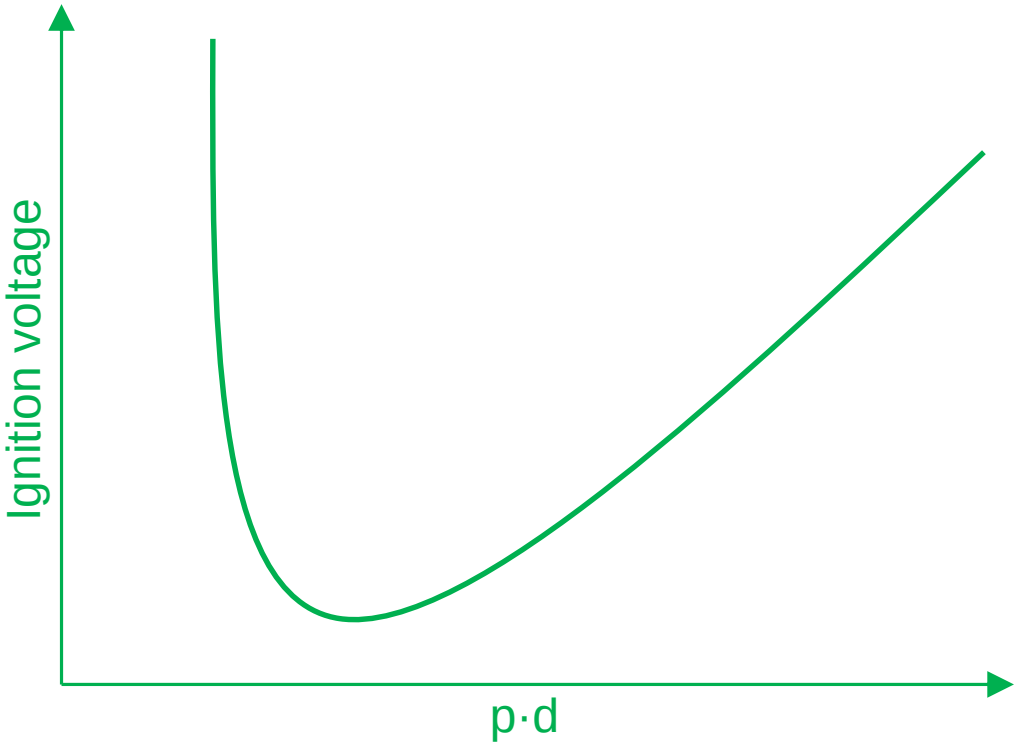
- 64. Why do we usually not consider the Townsend coefficient β?
  - It describes the probability of the generation of charge carriers by the collision of neutrals…which is negligible.

- 65. Give a short definition of the second Townsend coefficient.
  - γ is a material constant of an electrode material. It indicates how many electrons are released per incident ion.

- 66. Which conditions need to be fulfilled for a self-maintaining ignition of a discharge within a gas-filled gap?
  1. The electron energy must be sufficient for initiating ionisation of atoms or molecules (or even already ionised positive charge carriers).
  2. The initial electron of an avalanche must not be generated by an external source (The very first initial electron, however, is generated by such a source…)

- 67. Give a short definition of Paschen's law.
  - Paschen's law describes the dependency of voltage on the product of pressure and distance of electrodes, i.e., discharge gap thickness.

- 68. Draw a qualitative sketch of a Paschen's curve…do not forget to label both the y- and the x-axis.



- 69. What is the so-called ‘cathode fall’?
  - Within a plasma source, the main fraction of the applied voltage is found within a limited zone close to the cathode where – in comparison to other regions within the plasma source – a high electric field strength occurs. This zone is referred to as the cathode fall. Here, ions are efficiently accelerated towards the cathode surface and cause field emission of electrons.

- 70. What is the consequence of field emission of electrons caused by the energy transfer of incident ions?
  - The electrons emitted in this way can be accelerated on their way towards the anode and thus induce avalanche ionisation. Field emission is thus a mechanism for operating self-maintaining plasma discharges.

- 71. What is a ‘glow discharge’?
  - A glow discharge is a self-maintaining gas discharge between two electrodes.

- 72. A glow discharge comes along with the formation of characteristic dark and bright zones between the electrodes. Name at least three of these zones.
  - Aston dark space
  - Cathode glow
  - Cathode dark space,
  - Negative glow
  - Faraday dark space
  - Positive column
  - Anode glow
  - Anode dark space

- 73. Which type of discharge features a higher discharge current, the glow discharge or the anomalous glow discharge?
  - The latter one

- 74. What is the main difference of a Townsend discharge and a glow discharge?
  - Townsend discharge = non-self-maintaining discharge operated below the ignition voltage; glow discharge = a self-maintaining discharge (that is formed once the ignition voltage is reached).

- 75. Define ‘non-self-maintaining’ and ‘self-maintaining’.
  - Non-self-maintaining: generation of initial electron of avalanche by external source
  - Self-maintaining: generation of initial electron of avalanche by plasma species/effects (and not by external source)

- 76. What does the Child-Langmuir-law indicate?
  - $I \sim V^{3/2}$

- 77. When does thermionic emission of charge carriers occur?
  - The temperature of the cathode must be high enough so that a sufficient amount of electrons can cross the potential barrier between the valence band and the conductive band via the tunnel effect.

- 78. Which type of plasma is driven by thermionic emission?
  - The arc discharge.

- 79. Further increase in voltage of an anomalous glow discharge may lead to the formation of an arc discharge. What happens to the current density and the voltage in this case?
  - The current density within the plasma is increased whereas the voltage drops notably. Hence, the plasma channel features high conductivity.

- 80. Name at least two examples for arc discharges.
  - Flashes
  - Ignition plugs aka spark plugs
  - Plasma between the overhead contact line and the current collectors of trains, tramways and busses, see photo:

- 81. What is – approximately – the current density within an isothermal arc discharge?
  - about 1 kA/cm²

- 82. Which mechanism supports the (spatial) constriction of an arc plasma occur?
  - A geometrically favoured part of the cathode (e.g. a roughness peak) is preferentially heated, causing high thermionic emission of electrons and such a high local current density.

- 83. Give the typical temperature range of arc discharges.
  - about 4000 to 50 000K

- 84. What is the minimum temperature for thermionic emission of electrons?
  - It depends on the electrode material and pressure…
    - In vacuum: about 900K
    - At atmospheric pressure: about 3000K

- 85. Which equation can be applied for the determination of current density during thermionic emission?
  - The Richardson equation, aka Richardson-Dushman equation

- 86. What is the Schottky effect?
  - The effective work function of the cathode material is reduced by the high electric field strength at the cathode.

- 87. What is a plasma filament?
  - A filament is the rudiment of an arc, featuring very low diameters (some microns).